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SK6812 MINI-E RGB (similar to WS2812B) Pixel LED Chip 32228SMD Reverse Mount IC Individually Addressable Full Color DC5V

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Reviews | 4.9 ★★★★★ ☆ 21 ratings ✓ All from verified purchases



★★★★★

Wattage: 200PCS

Perfect and in good condition. They are exactly the LEDs I need for my project.



E***s | October 20, 2024

👍

It has helped you (0) ...



★★★★★

Wattage: 50PCS

As advertised. Thank you.

S***t | October 24, 2024

👍

It has helped you (0) ...



★★★★☆

Wattage: 50PCS

Some LEDs were short and 1 had a broken pin, but I ordered 50, they all work as expected

Anonymous | January 25, 2025

👍

It has helped you (0) ...

See more

Details

Luminous flux	<60	Very concerned chemical	None
Smart device	FORKS	Power supply	SWITCH
Model number	WS2812B	LED chip model	SMD5050
Half-life (hours)	More than 50,000 hours	Certification	CCC,CE,FCC,ROHS
Energy source	DC	Origin	CN(Origin)
Strip type	SMD	Power consumption (W/m)	2.88 W/m
Chance	Lounge	Specifications (light points per meter)	3228
Item type	Band	Brand name	LOAMLIN
Voltage	DC5V	Number of LEDs / M	NA
LED chip brand	Epistar	Waterproof	No
Color temperature (K)	2200-6500K		

SK6812MINI-E

3.2x2.8x1.78 mm 0.2W Intelligent external control surface mount SMD LED (MSL:5a)



Valoraciones (21) Detalles  **Descripción** Tienda Te podría interesar



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SK6812MINI-E is a smart LED control circuit and light emitting circuit in one controlled LED source, which has the shape of a 3528 LED chip. Each lighting element is a pixel, and the intensities of the pixels are contained within the intelligent digital interface input. The output is driven by patented PWM technology, which effectively guarantees high consistency of the color of the pixels. The control circuit consists of a signal shaping amplification circuit, a built-in constant current circuit, and a high precision RC oscillator.

The data protocol being used is unipolar RZ communication mode. The 24-bit data is transmitted from the controller to DIN of the first element, and if it is accepted it is extracted pixel to pixel. After an internal data latch, the remaining data is passed through the internal amplification circuit and sent out on the DO port to the remaining pixels. The pixel is reset after the end of DIN. Using automatic shaping forwarding technology makes the number of cascaded pixels without signal transmission only limited by signal transmission speed.

The LED has a low driving voltage (which allows for environmental protection and energy saving), high brightness, scattering angle, good consistency, low power, and long life. The control circuit is integrated in the LED above.

2. Main Application Field:

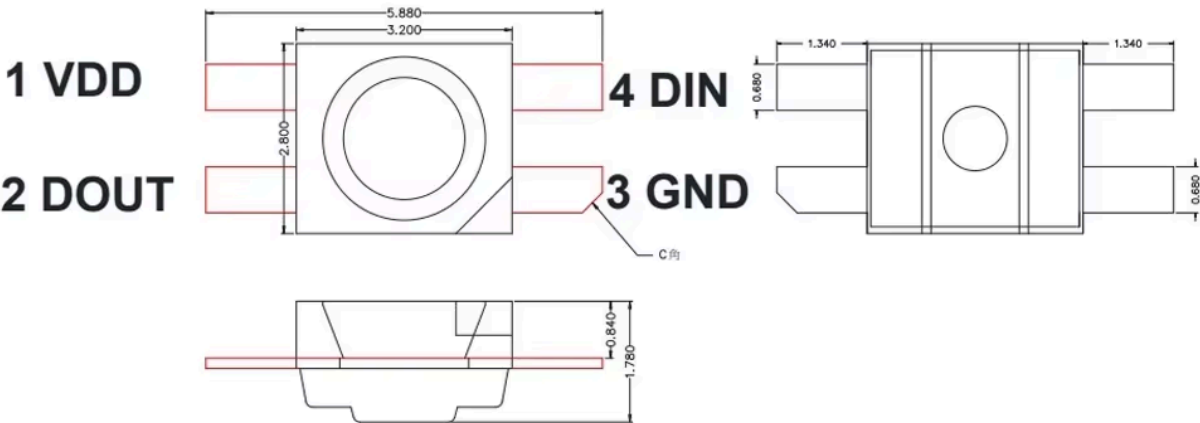
- Full color LED string light, LED full color module, LED super hard and soft lights, LED guardrail tube, LED appearance / scene lighting
- LED point light, LED pixel screen, LED shaped screen, a variety of electronic products, electrical equipment etc..

3. Description:

- Top SMD internal integrated high quality external control line serial cascade constant current IC;
- control circuit and the RGB chip in SMD 3528 components, to form a complete control of pixel, color mixing uniformity and consistency;
- built-in data shaping circuit, a pixel signal is received after wave shaping and output waveform distortion will not guarantee a line;
- The built-in power on reset and reset circuit, the power does not work;
- gray level adjusting circuit (256 level gray scale adjustable);
- red drive special treatment, color balance;
- line data transmission;
- plastic forward strengthening technology, the transmission distance between two points over 10M;
- Using a typical data transmission frequency of 800 Kbps, when the refresh rate of 30 frames per sec



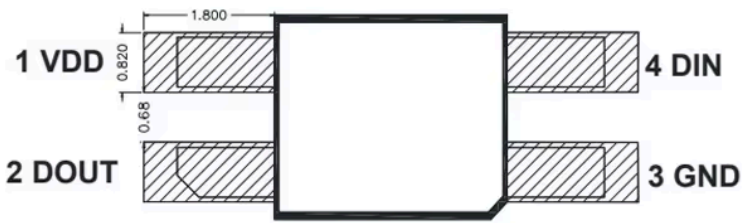
Mechanical Dimensions:



- Notes:
- 1. All dimensions are in millimeters.
 - 2. Tolerance is $\pm 0.1\text{mm}$ unless otherwise noted

5. PIN configuration

NO.	Symbol	Function description
1	VDD	Power supply LED
2	DOUT	Control data signal output
3	GND	Ground
4	DIN	Control data signal input



7. General description of product naming

SK

6812

MINI-E

①

②

③

①	②	③
Series	IC series and current code	package outline
The default is to integrate the RGB chip with the IC in the	Refers to the 68 series IC 5/12MA current version	3.2x2.8x1.78mm package outline

8. Electrical parameters（Ta=25℃,VSS=0V）：

Parameter	Symbol	Range	Unit
Power supply voltage	VDD	+3.7~+5.5	V
Logic input voltage	V _{IN}	-0.5~VDD+0.5	V
Working temperature	T _{opt}	-40~+85	℃
Storage temperature	T _{stg}	-50~+150	℃
ESD pressure(HBM)	V _{ESD}	4K	V
ESD pressure(DM)	V _{ESD}	200	V



Color	SK6805MINI -E 5mA			SK6812MINI -E 12mA		
	Dominate Wavelength(nm)	Luminance(mcd)	luminous flux(lm)	Dominate Wavelength(nm)	Luminance(mcd)	luminous flux(lm)
红色 (RED)	620-630	100-200	0.5-1.0	620-625	400-700	1.0-2.0
绿色 (GREEN)	520-535	400-700	2.0-3.0	520-530	1000-1500	3.0-4.0
蓝色 (BLUE)	460-475	50-100	0.1-0.5	460-470	200-400	0.5-1.0

10. The IC electrical parameters (unless otherwise specified, TA=-20 ~ +70 °C, VDD=4.5 ~ 5.5V, VSS=0V):

Parmeter	Symb ol	Min	Typic al	Max	Unit	Test conditions
The chip supply voltage	VDD	---	5.2	---	V	---
The signal input flip threshold	VIH	0.7*VDD	---	---	V	VDD=5.0V
	VIL	---	---	0.3*VDD	V	
The frequency of PWM	FPWM	---	1.2	---	KHZ	---
Static power consumption	IDD	---	1	---	mA	---

11. Switching characteristics(VCC=5V Ta=25 °C):

Parameter	Symbol	Min	Typical	Max	Unit	Test conditions
The speed of data transmission	fDIN	---	800	---	KHZ	The duty ratio of 67% (data 1)
DOUT transmission delay	TPLH	---	---	500	ns	DIN→DOUT
	TPHL	---	---	500	ns	
IOUT Rise/Drop Time	Tr	---	100	---	ns	VDS=1.5 IOUT=13mA
	Tf	---	100	---	ns	

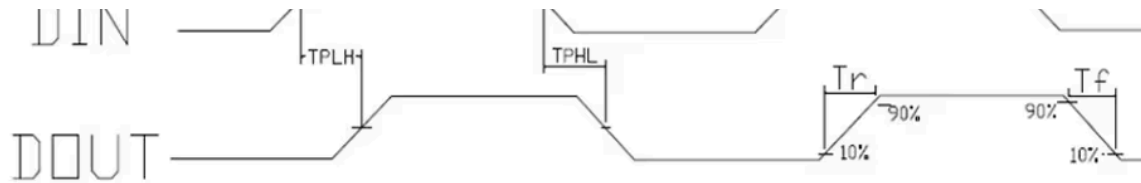


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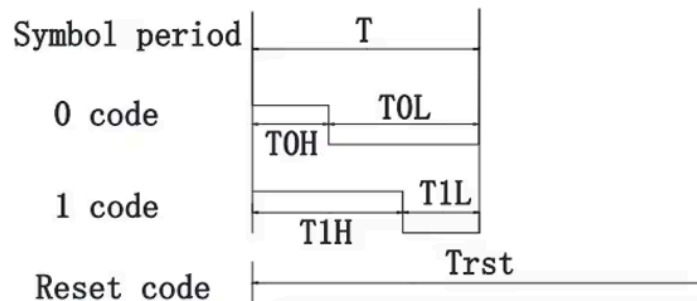
12. The data transmission time :

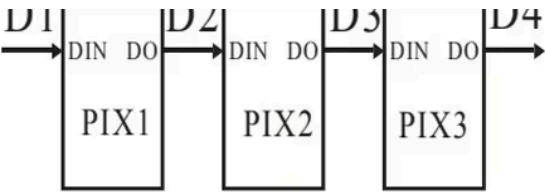
Name		Min.	Standard value	Max.	Unit
T	Code period	1.20	--	--	μs
T0H	0 code, high level time	0.2	0.32	0.4	μs
T0L	0 code, low level time	0.8	--	--	μs
T1H	1 code, high level time	0.58	0.64	1.0	μs
T1L	1 code, low level time	0.2	--	--	μs
Trst	Reset code, low level time	>80	--	--	μs

1. The protocol uses a unipolar zeroing code. Each symbol must have a low level. Each symbol in this protocol starts with a high level. The high time width determines the "0" or "1" code. .
2. When writing programs, the minimum symbol period is $1.2 \mu\text{s}$.
3. The high time of "0" code and "1" code should be in accordance with the stipulated range in the above table. The low time requirement of "0" code and "1" code is less than $20 \mu\text{s}$.

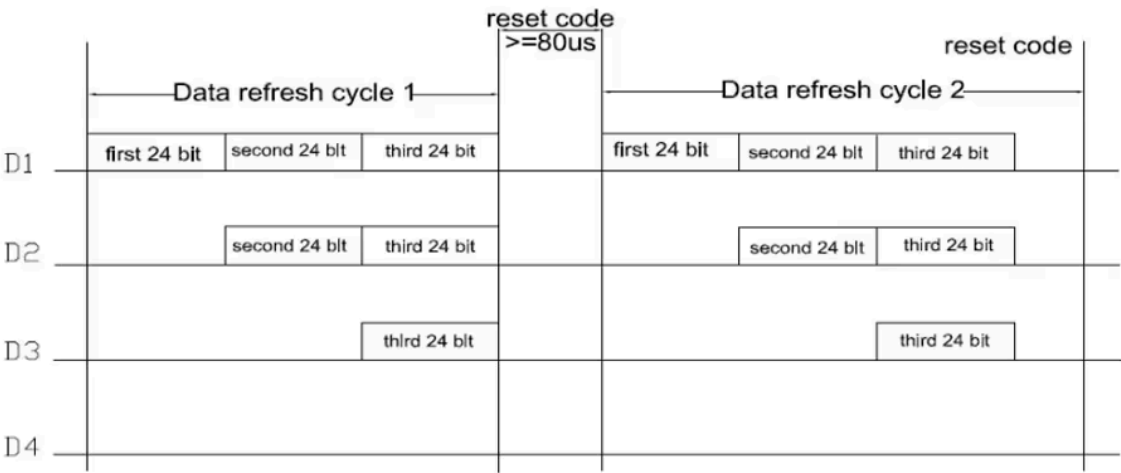
13. Timing waveform:

Input code:

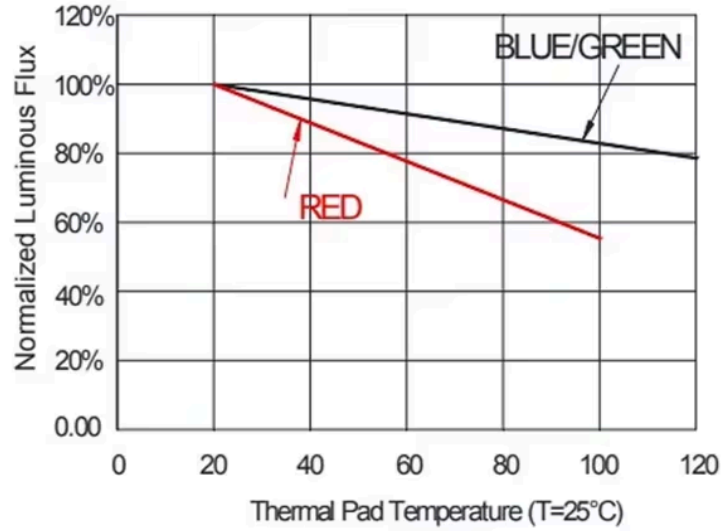




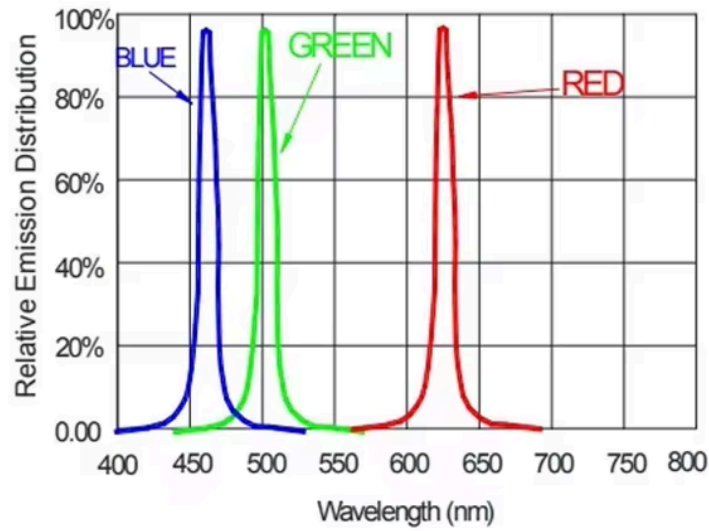
14. The method of data transmission:



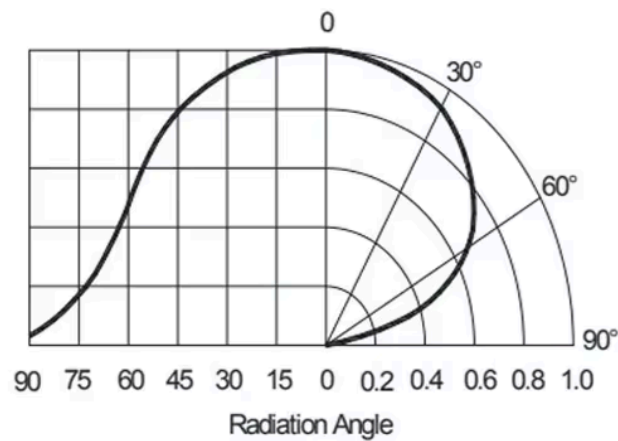
Note: the D1 sends data for MCU, D2, D3, D4 for data forwarding automatic shaping cascade circuit.



Wavelength Characteristics



Typical Radiation Pattern 120°





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Application Case



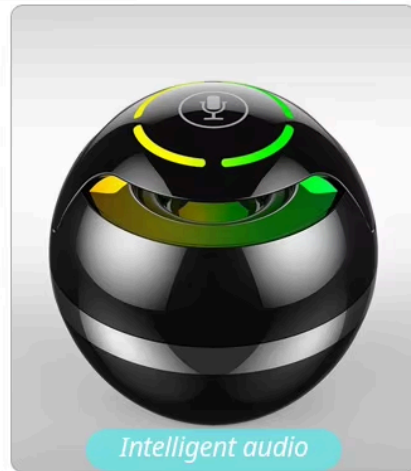
Mechanical keyboard



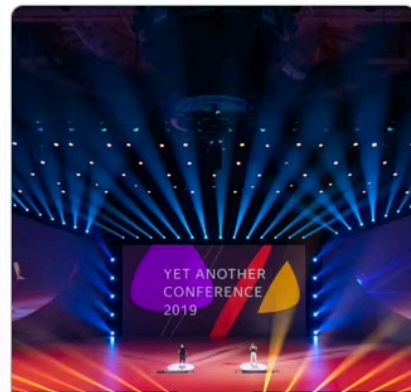
Glitzy shoes



Intelligent case and bag



Intelligent audio





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Product certification



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Communication



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